TLS

TRAINING NOTES

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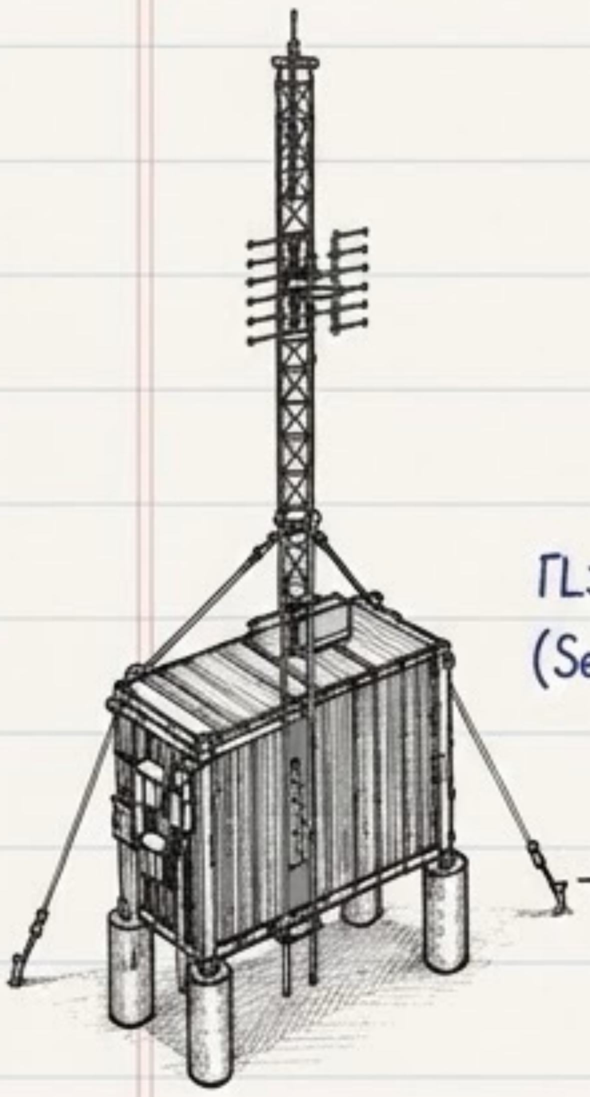
HOW TLS WORKS ?

TLS
TRAINING NOTES

TLS (Transponder Landing System) is a ground based precision approach system.
It provides dynamic Localizer and Glide Slope guidance to the aircraft.

Aircraft Transponder (Replies)



TLS Interrogator (Sends Signal)

→ Transmit → Target → Reply → Receive

KEY IDEA

TLS interrogates the aircraft transponder, measures the reply and calculates the aircraft position in 3D.

TLS MEASURES:

- Range (Distance)
- Azimuth (Direction)
- Elevation (Height)

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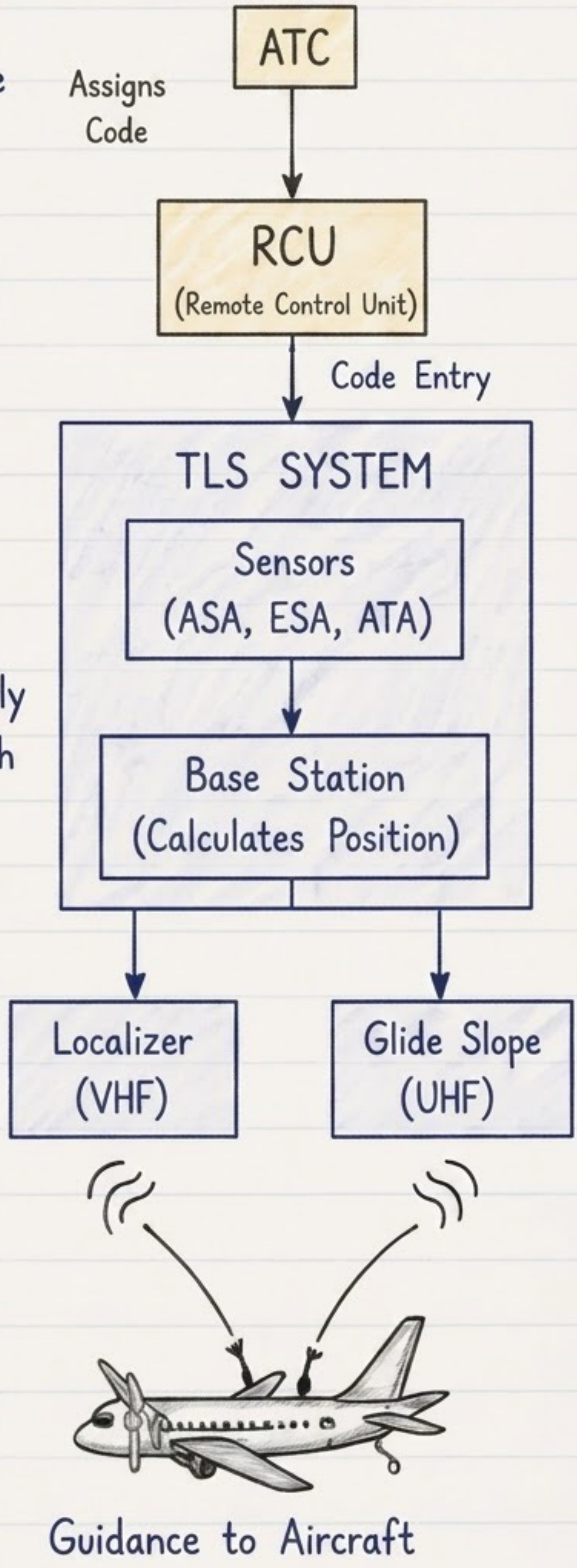
1. OPERATION OVERVIEW

1. ATC assigns a 4-digit code to the aircraft.
2. The code is entered into the TLS (RCU).
3. TLS interrogator sends a radio signal (1090 MHz).
4. Aircraft transponder replies with the same code.
5. TLS sensors receive the reply and measure Range, Azimuth & Elevation.
6. The Base Station calculates the aircraft position.
7. TLS generates Localizer & Glide Slope guidance.
8. Guidance is transmitted to the aircraft (VHF / UHF).
9. The aircraft receives the signals and the pilat follows the guidance to the runway.



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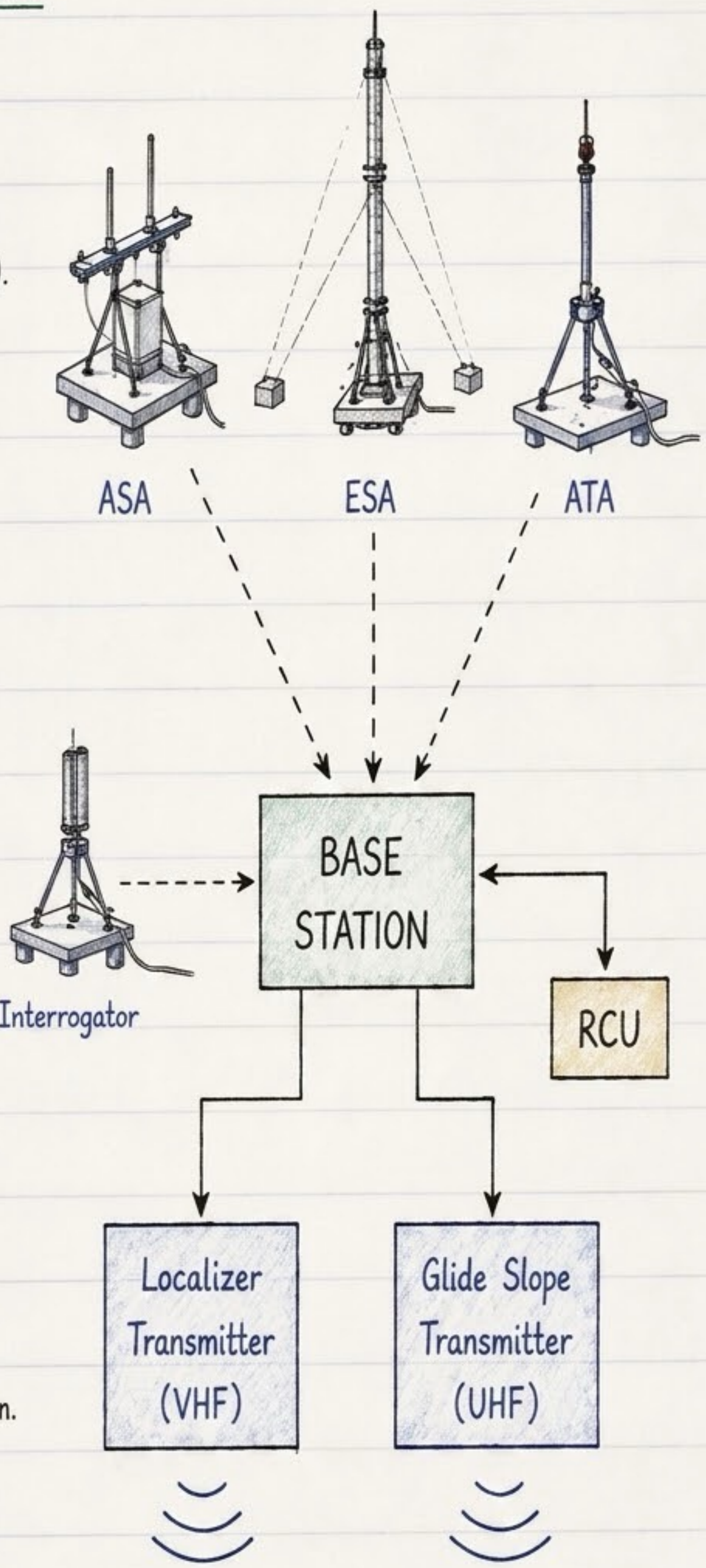
2. TLS COMPONENTS



ASA

Azimuth Sensor Array

Measures azimuth (direction).

ESA

Elevation Sensor Array

Measures elevation (height).

ATA

Alternate Time of Arrival

Provides time reference.

Interrogator

Sends interrogation signal to the aircraft.

Base Station

Processes measurements and calculates the aircraft position.

RCU

Remote Control Unit used to control and monitor the system.

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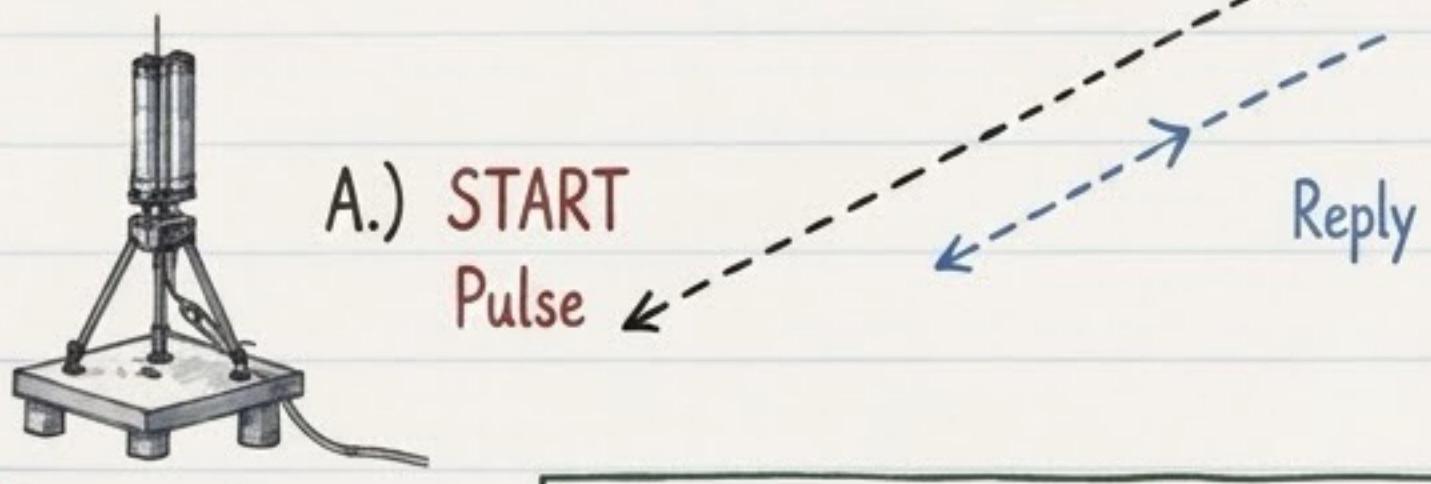
3. HOW POSITION IS CALCULATED

RANGE (Distance)

Interrogator sends a START pulse.
Sensors measure Time Of Arrival (TOA)
of the aircraft reply.

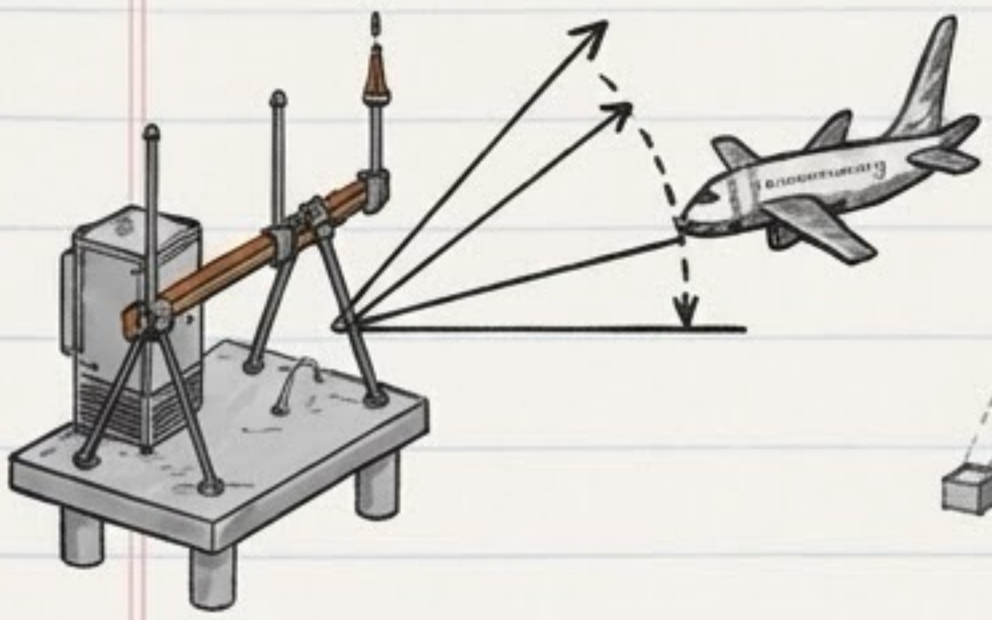


Interrogator

$$\text{Distance} = \frac{\text{Speed of Radio Wave} \times \text{Time}}{2}$$

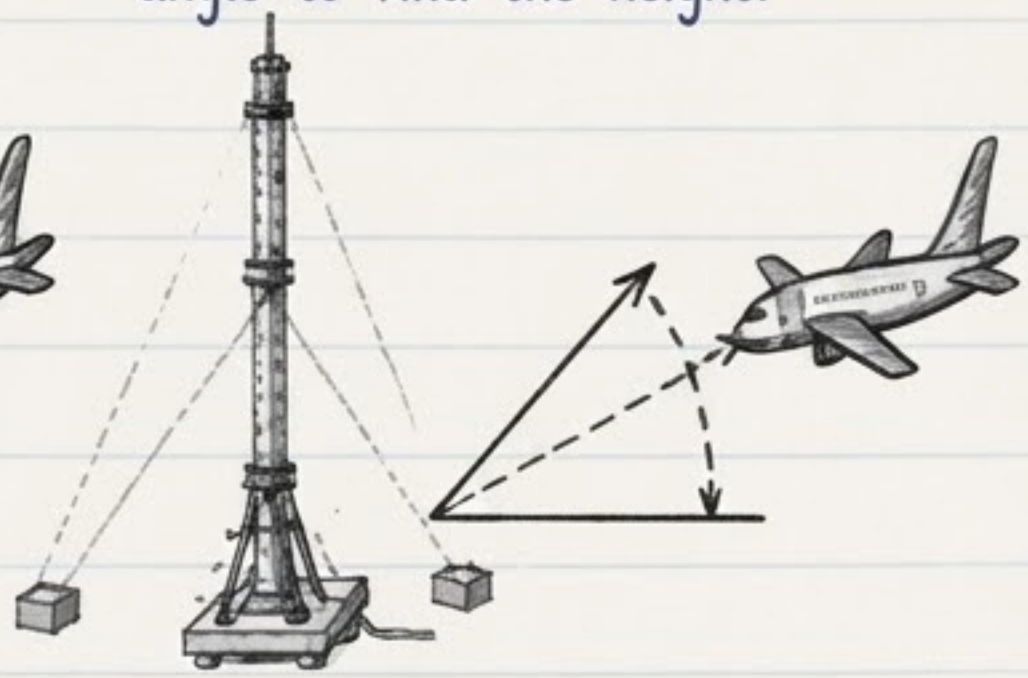
AZIMUTH (Direction)

ASA measures the Angle Of Ar AOA to find the airraft direction.



ELEVATION (Height)

ESA measures the elevation angle to find the height.



ESA

These three measurements give the aircraft 3D position.

[Range, Azimuth, Elevation]

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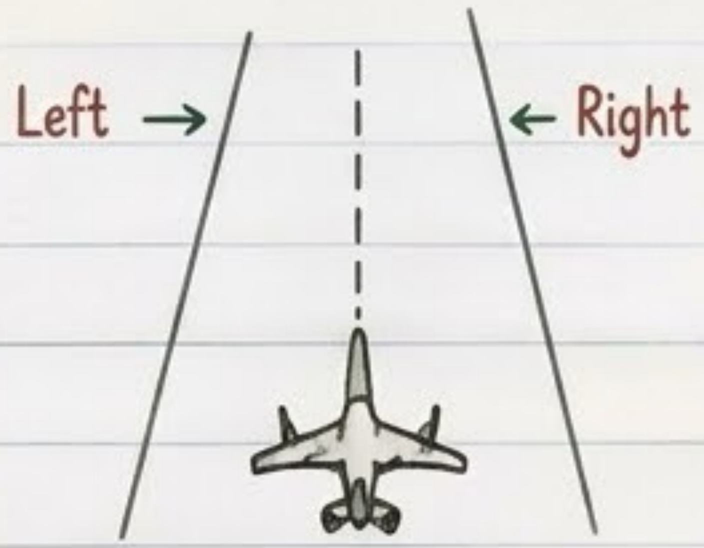
HOW TLS WORKS ?

4. GUIDANCE GENERATION

The Base Station compares the aircraft position with the desired approach path. It calculates the deviation and creates Localizer and Glide Slope guidance.

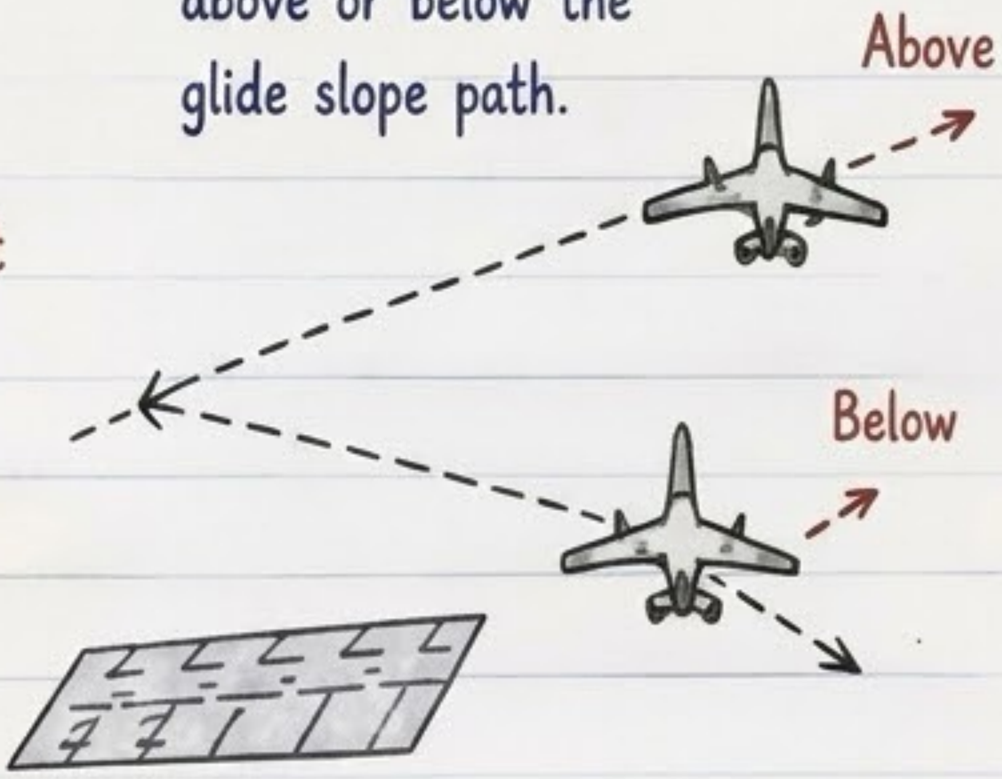
LOCALIZER (Left / Right)

Tells the aircraft if it is left or right of the centerline.

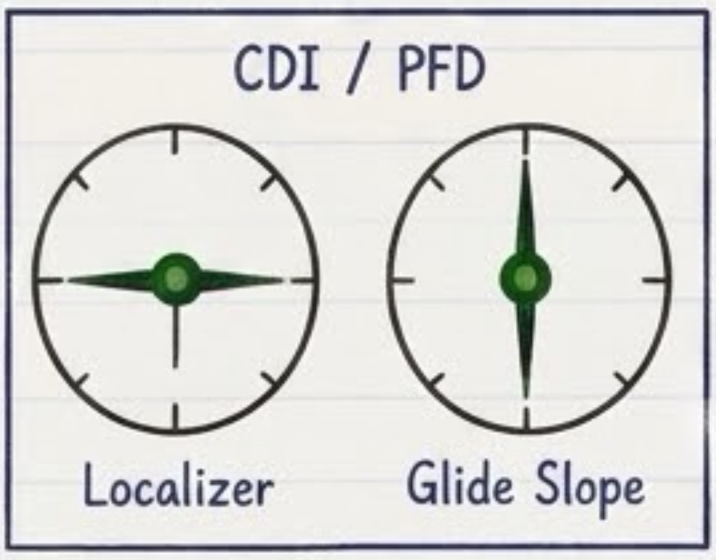


GLIDE SLOPE (Up / Down)

Tells the aircraft if it is above or below the glide slope path.



When both needles are at center (●), the aircraft is on the correct path. Follow the guidance to land safely.



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5. SIGNALS & FREQUENCIES

INTERROGATION

Interrogator → Aircraft

Frequency: 1090 MHz

REPLY

Aircraft Transponder → TLS Sensors
(With aircraft code)

Frequency: 1090 MHz

LOCALIZER SIGNAL

TLS → Aircraft

Frequency: VHF
(108 - 118 MHz)

GLIDE SLOPE SIGNAL

TLS → Aircraft

Frequency: UHF
(329 - 335 MHz)

UPDATE RATE

TLS updates the aircraft position and guidance about 10 times per second (10 Hz).

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6. KEY POINTS

- Tracks only the selected aircraft by its 4-digit code.
- If two aircraft use the same code, TLS will not provide guidance.
- Provides dynamic guidance like ILS, bassed as a pooke of is but based on transponder.
- Works day and night, in all weather.
- Uses radio waves, not light.
- Stops guidance when aircraft is above ~7° elevation or outside service volume.

☆ TLS = Transponder Landing System

A ground based precision approach system.

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